



A Risk Assessment Matrix for Identifying Likelihood of Loss to Follow Up in Persons Living with HIV



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Background

Retention in HIV care is essential to achieving the UNAIDS 95-95-95 targets and sustaining viral suppression at the population level. For the Southern Regional Health Authority (SRHA), in Jamaica, Lost to Follow Up (LTFU) is a persistent issue faced by clients within the region. Continuous engagement in care supports sustained antiretroviral therapy (ART) adherence, leading to improved virologic suppression and reduced morbidity and mortality. Conversely, patients who disengage from care face increased risks of virologic failure, and disease progression. These individual-level consequences can undermine broader public health goals, particularly efforts to control HIV transmission. As at June 2024, the SRHA had 2566 patients on register, 2074 retained in care (81%) and 492 clients lost to follow-up (19%). Our study looked at the development of a risk assessment matrix that determined a patient's risk of becoming Lost To Follow Up.

Introduction

Globally, LTFU remains a major barrier to achieving the UNAIDS 95-95-95 targets, especially in low- and middle-income countries where structural, social, and clinical barriers often converge to hinder retention. Jamaica’s Southern Regional Health Authority (SRHA) continues to face persistent LTFU challenges, which demand innovative, data-driven solutions to identify and support at-risk clients. Risk assessment matrices have emerged as effective tools for prioritizing patient interventions by systematically evaluating multiple factors associated with disengagement from care. When applied to HIV programs, these matrices can help clinicians and program managers allocate resources more effectively and design targeted retention strategies.

The new scoring system is meant to separate patients into different risk groups based on a number of health and engagement-related characteristics. On a scale from 0 to 100, each element gets a score. Higher scores mean that the person is more vulnerable or at danger. The factors considered in the computation are: length of time in care, frequency of loss to follow-up (LTFU), address type, last location visited (by parish), length of time lost to follow-up, gender, connection to care, and viral suppression status. The overall score is made up of these factors, each with a different weight: length of time in care (20%), frequency of LTFU (20%), address (10%), last site visited (10%), length of time lost to follow-up (10%), gender (10%), connection to care (10%), and suppression status (10%). The weighted total of these criteria gives the overall risk score.

To ensure the objectivity and statistical validity of such tools, analytic methods such as the Chi-square test are often employed to examine associations between key predictors and LTFU outcomes. This combination of structured risk assessment and statistical analysis enhances decision-making, enabling health systems to address retention challenges more strategically and effectively.

Objective

To develop a risk assessment matrix that determines the risk level of a patient becoming Lost To Follow Up, that is, rating factors that can negatively impact adherence to care and assigning overall scores to identify clients who are likely to miss their appointment date by at least ninety days.

Methods

A risk assessment scoring matrix was created to determine the risk level of HIV patients becoming LTFU within SRHA treatment sites in Jamaica as of June 2024. The retention risk predictors selected were: length of time in care, frequency of loss to follow-up, address, length of time lost to follow-up, last site visited, last viral load status, time to linkage, and gender. For each retention risk matrix predictor, a set of criteria were defined and assigned points ranging from 0-100 points to create the scoring matrix. The Scoring matrix was then used to assess and score patients divided into two groups: retained in care and lost to follow-up as at the end of June 2024. A formula for weighted sum of all scores was developed to categorize the patient based on the final risk score. The following predictors were weighted: length of time in care (20%), frequency of loss to follow-up (10%), address (20%), length of time lost to follow-up (10%) last site visited (10%), last viral load status (10%), time to linkage (10%), and gender (10%). Patients were then categorized in the overall risk scoring scale as follows: 0-25.9 Very Low Risk, 26-49.9 Low Risk, 50-74.9 was Moderate Risk, and 75-100 was High to Very High Risk.

Data representing the Southern region was extracted from the National Treatment Site Information System, TSIS 2.0 at the end of June 2024, a total of 2615 client records were found. The inclusion criteria for this study was the records of clients attaining 18 years and older at the time of the data extraction; who were originally linked to a treatment site in the Southern health region and receiving ARVs for a minimum of 5 years. After this cleaning, a total of 2399 client records remained. Frequencies, Chi-Square and Phi and Cramer’s V analyses were done on the factors identified to compare categorical variables i.e. risk predictors (includes overall risk scores) and determine their association with a client becoming Lost to Follow Up. The analyses were done on the 26th version of the Statistical Package for the Social Sciences (SPSS) software.

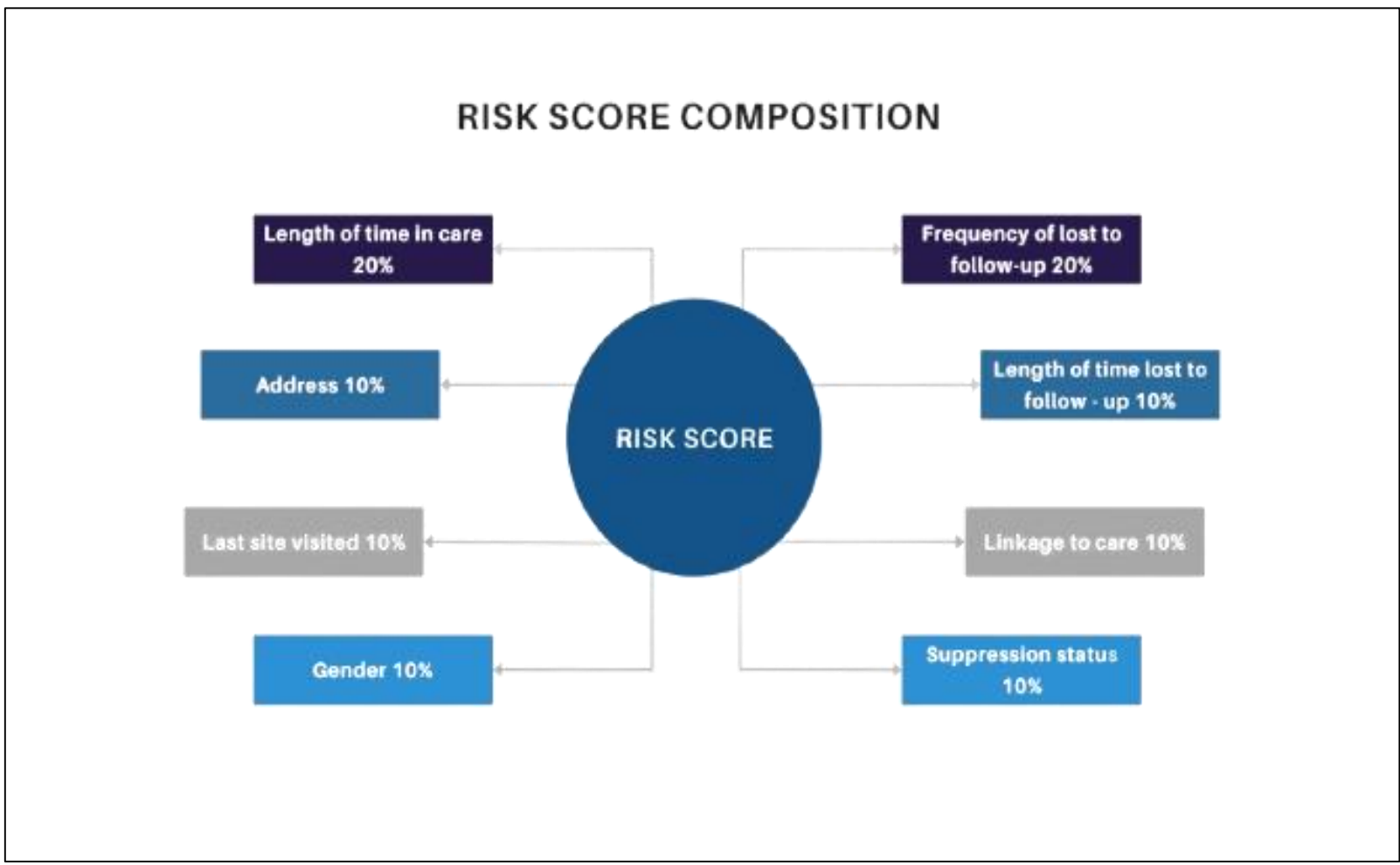


Figure 1. Relationship Chart Depicting Risk Score Composition

Table 1. Showing Retention Risk Predictors and Risk Assessment Scoring Criteria

Durati on in Care	Frequen cy of LTFU	Address Type	Last Site Visited	Length of Time LTFU	Last Suppressio n Status	Linkage to Care	Gender	Score Assigned	Risk Level
20%	20%	10%	10%	10%	10%	10%	10%	100%	
0-1 year (0-12 months)	7+ Times	Rural & Not Available	Northern & Western Sites (e.g., St. Mary, Portland, Trelawny, St. James, Hanover, Westmorela nd)	12+ Months	Unsuppress ed	>112 days (more than 16 weeks)	Male* (60 points)	100 points	Very High Risk
1-2 years (13-24 months)	5-6 Times			6-12 months		85-112 days (5-8 weeks)	Male* (60 points)	75 points	High Risk
2-3 years (25-36 months)	3-4 Times	Structured Communiti es	Southeaster n Sites (St. Catherine, Kingston, St. Andrew, St. Thomas)	3-6 months		57-84 days (9-12 weeks)	Female* (40 points)	50 points	Moderat e Risk
3-4 years (37-48 months)	1-2 Times			1-3 months		29-56 days (13-16 weeks)		25 points	Low Risk
4+ years (49+ months)	0 Times	Developed	Southern Sites (Manchester Clarendon, St. Elizabeth)	0-1 months	Suppressed	0-28 days (0-4 weeks)		0 points	Low/No Risk

*Scoring done based on the percentage of males and females in the data set and health seeking behaviour tendency.

The overall score is the weighted sum of the scores for each factor:

Overall Score=0.2×(Score for Length of Time in Care)+0.2×(Score for Frequency of Loss to Follow-Up) +0.1×(Score for Address) +0.1×(Score for Last Site Visited) +0.1×(Score for Length of Time Lost to Follow-Up) +0.1×(Score for Gender) +0.1x (Score for Linkage to Care) + 0.1x (Score for Suppression Status)

Figure 2. Calculation of Overall Scores

Table 1.1 Showing Final Scoring Categorization of Patient’s Retention Risk Level

Category	Score Range	Overall Risk Score Level	Description	Likely Client Characteristics
Category 1	0 – 25.9	Very Low Risk	Clients are stable in care with minimal risk of disengagement.	Recently linked to care, consistently virally suppressed, frequent follow-ups, stable address.
Category 2	26 – 49.9	Low Risk	Clients may have minor issues that require light monitoring.	Occasional missed visits, minor delays in linkage, and some instability in address or site transfers.
Category 3	50-74.9	Moderate risk	Clients show early warning signs of becoming LTFU.	Irregular visit history, longer time to linkage, inconsistent suppression, or moderate relocation.
Category 4	75-100	High- Very High Risk	Clients are disengaged from some parts of the care continuum and need targeted interventions. OR Clients are most likely to be LTFU and need urgent re-engagement strategies.	Prolonged time lost to follow-up, unsuppressed, delayed linkage, multiple site changes. OR Unsuppressed, >16 weeks delay in linkage, long time LTFU, unstable housing, or site abandonment.

Results and Conclusions

The frequency analysis done on the data set found that the majority of clients who met the inclusion criteria were female (58.6%, n=1407) and that the highest percentage of clients were 50 years and older(36.8%, n=882). More than half the clients lived in rural areas (54.2%, n=1302) and almost a quarter of the sample had no address listed (23.3%, n=558). Finally, the majority of clients reviewed were retained in care at the time of data extraction.

Table 2.0 Demographics of Persons living with HIV on register within Treatment Sites in the Southern Region, Jamaica 2024. (N=2399)

Sex at Birth			
	Frequency	Percent	
Female	1407	58.6	
Male	992	41.4	
Total	2399	100	
Age Group			
	Frequency	Percent	
15-19	15	0.6	
20-24	97	4	
25-29	175	7.3	
30-34	245	10.2	
35-39	307	12.8	
40-44	340	14.2	
45-49	338	14.1	
50 and over	882	36.8	
Total	2399	100	
Address Type			
	Frequency	Percent	
Developed	60	2.5	
Not Available	558	23.3	
Rural	1302	54.2	
Structured Community	479	20	
Total	2399	100	
Retained in Care			
	Frequency	Percent	
No	462	19.3	
Yes	1937	80.7	
Total	2399	100	
Risk Score Categorized			
	Lost To Follow Up	Retained in Care	Total
Low Risk	0	20	20
Moderate Risk	128	740	868
High Risk	222	1035	1257
Critical Risk	90	58	148
Total	440	1853	

Majority (80%) of the eight (8) predictors contributed to the overall score statistical significance, p <0.001. Patients with Low and Moderate risk scores showed very high retention (100% and 85% retained, respectively), while those in the Critical Risk category have the highest LTFU (61% lost). Gender and linkage time had no statistically significant differences. For the factors that had a Cramer’s V result, it was found that the time spent in care(0.317), and the frequency of LTFU status (0.453) had higher association strengths than a client’s age at registration (0.226) and their last viral load status (0.266). The weakest associations were seen in gender (0.036), diagnosis year (0.170), home parish (0.135), address type (0.130), home parish (0.135), and last site visited (0.164).

Table 2.1 Key Observations from the study and their Statistical Significance

Factor	Statistical Significance	Strength of Association	Key Findings
Gender	Not significant (P=0.075)	Weak (0.036)	No meaningful difference in retention between males and females.
Age at Registration	Significant (P<0.001)	Moderate (0.226)	Retention increases with age: 55+ (91%) vs 18–24 (73%).
Current Age Group	Significant (P<0.001)	Weak (0.167)	18–24: 67% retained, 33% LTFU vs 55+: 89% retained.
Diagnosis Year	Significant (P<0.001)	Weak (0.170)	Pre-1999: 91% retained vs 2010–2019: 23% LTFU.
Linkage Year	Significant (P<0.001)		Greater retention for earlier linkage years (e.g., 88% retention in 2000-2009) compared to recent linkage years (2020+, with 18% lost to follow-up).
Home Address (Parish)	Significant (P<0.001)	Weak (0.135)	High Retention: Manchester (86%), Clarendon (83%). Low Retention: Westmoreland (57%), Hanover (100% LTFU).
Address Type	Significant (P<0.001)	Weak (0.130)	Developed Communities: 95% retained; Structured Communities: 87%; Rural Communities: 81%.
Last Viral Load Suppression	Significant (P<0.001)	Moderate (0.266)	Suppressed viral loads are strongly associated with retention (90% retained), while Unsuppressed viral loads are linked with greater loss to follow-up (35%).
Time in Care	Significant (P<0.001)	Moderate (0.317)	≥4 years: 88% retained vs 0–1 year: 50% retained. Retention improves with greater time in care.
LTFU Visit Frequency	Significant (P<0.001)	Moderate (0.453)	≥7 visits: 94% retained vs 0 visits: 49% retained. Retention improves with frequent visits.
Last Site Visited (Parish)	Significant (P<0.001)	Weak (0.164)	High Retention: Manchester (84%), Clarendon (80%). Poor Retention: Hanover (100% LTFU).
Linkage (Time to Linkage)	Not significant (P=0.420)		No retention differences by time to linkage.
Risk Score (Categorized)	Significant (P<0.001)		Patients with Low and Moderate risk scores show very high retention (100% and 85% retained, respectively), while those in the Critical Risk category have the highest loss to follow-up (61% lost).
Length of Time LTFU (Score)	Significant (P<0.001)		>12 months LTFU: 50% loss, less likely to return.

Conclusion: These results suggest that for better retention, lower risk scores were key. There were statistically significant differences in retention in care depending on factors such as age, address type, Frequency of loss to follow up, viral load status and length of time in care. Therefore the higher the risk score, the greater the likelihood of becoming loss to follow-up.